Topographical & Forensic Map Analysis: Sun Peaks Area

Application of spatial profiling and remote sensing in the disappearance of Ryan Shtuka.

SUBJECT FILE: RS-021718
COORDINATES: 50.884° N, 119.882° W

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Executive Summary: The Investigative Objective

> **Objective:** Apply advanced remote sensing principles to investigate topographical anomalies in the Sun Peaks terrain.



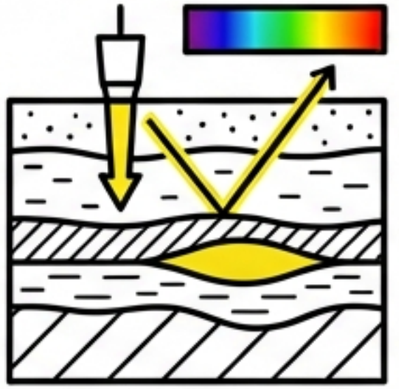
Phase 1: Geographic Profiling.
Establishing statistical search radiuses based on the Point Last Seen (PLS).

`Search Area: PLS + 3σ Radius`



Phase 2: Light Detection and Ranging (LiDAR).
Identifying localized elevation changes through dense canopy cover.

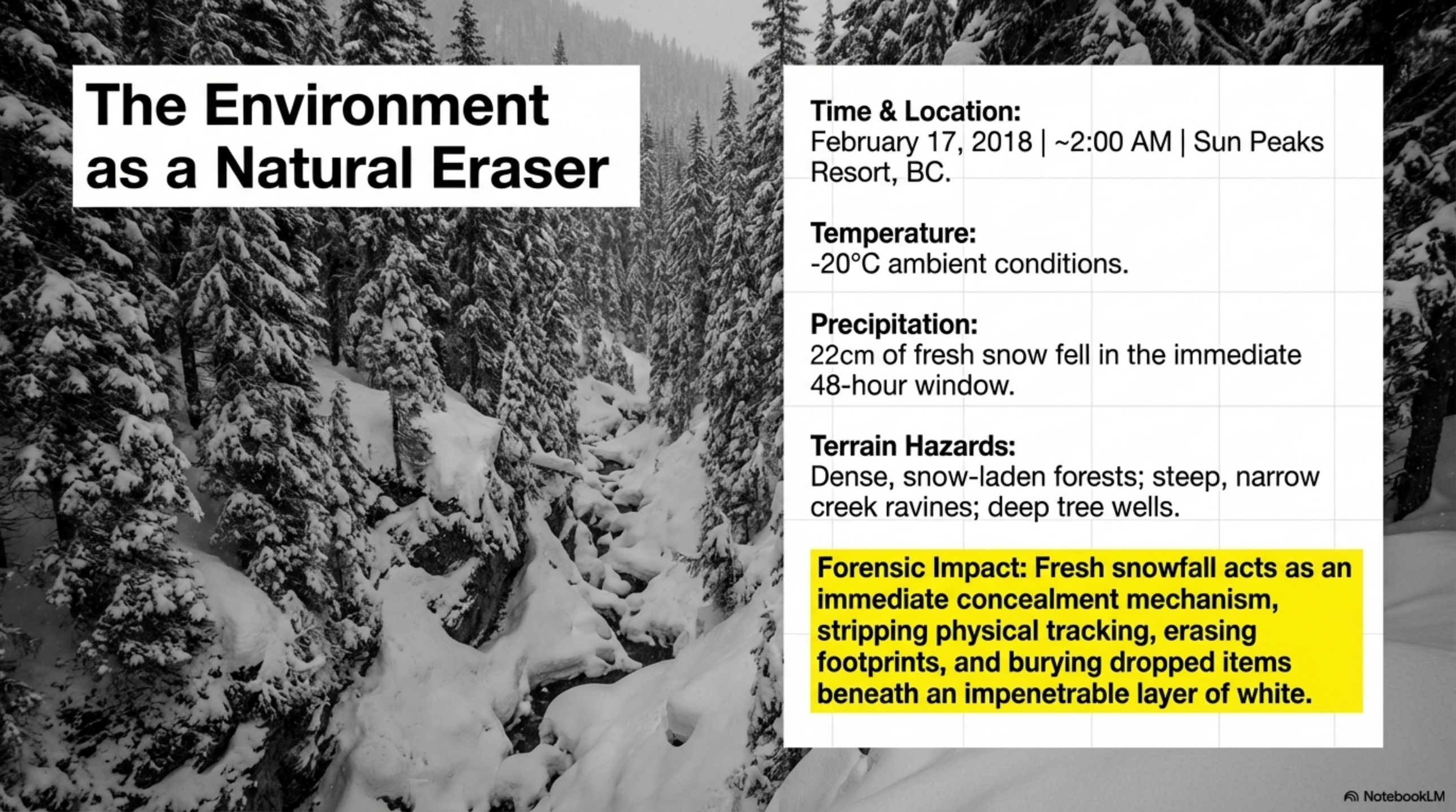
`Resolution: <0.5m Elevation Delta`



Phase 3: Near-Surface Geophysics.
Deploying Ground Penetrating Radar (GPR) and hyperspectral imaging for targeted anomaly resolution.

`GPR Depth: 10m Max Penetration`

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The Environment as a Natural Eraser

Time & Location:
February 17, 2018 | ~2:00 AM | Sun Peaks Resort, BC.

Temperature:
-20°C ambient conditions.

Precipitation:
22cm of fresh snow fell in the immediate 48-hour window.

Terrain Hazards:
Dense, snow-laden forests; steep, narrow creek ravines; deep tree wells.

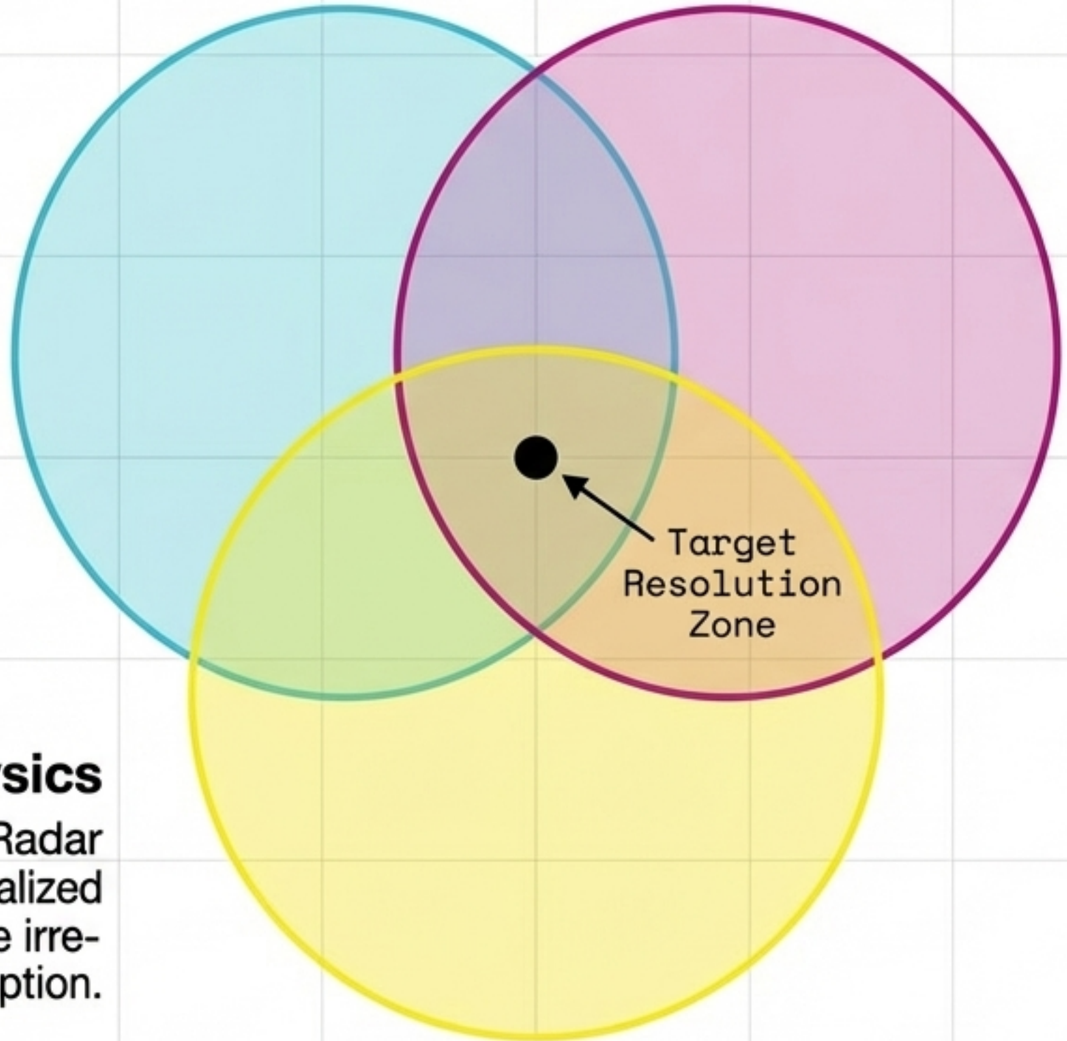
Forensic Impact: Fresh snowfall acts as an immediate concealment mechanism, stripping physical tracking, erasing footprints, and burying dropped items beneath an impenetrable layer of white.

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Multidisciplinary Search Methodology

1. Geographic Profiling

Narrows the macro-area using behavioral statistics of lost persons in mountainous terrain.



2. LiDAR / Hyperspectral Imaging

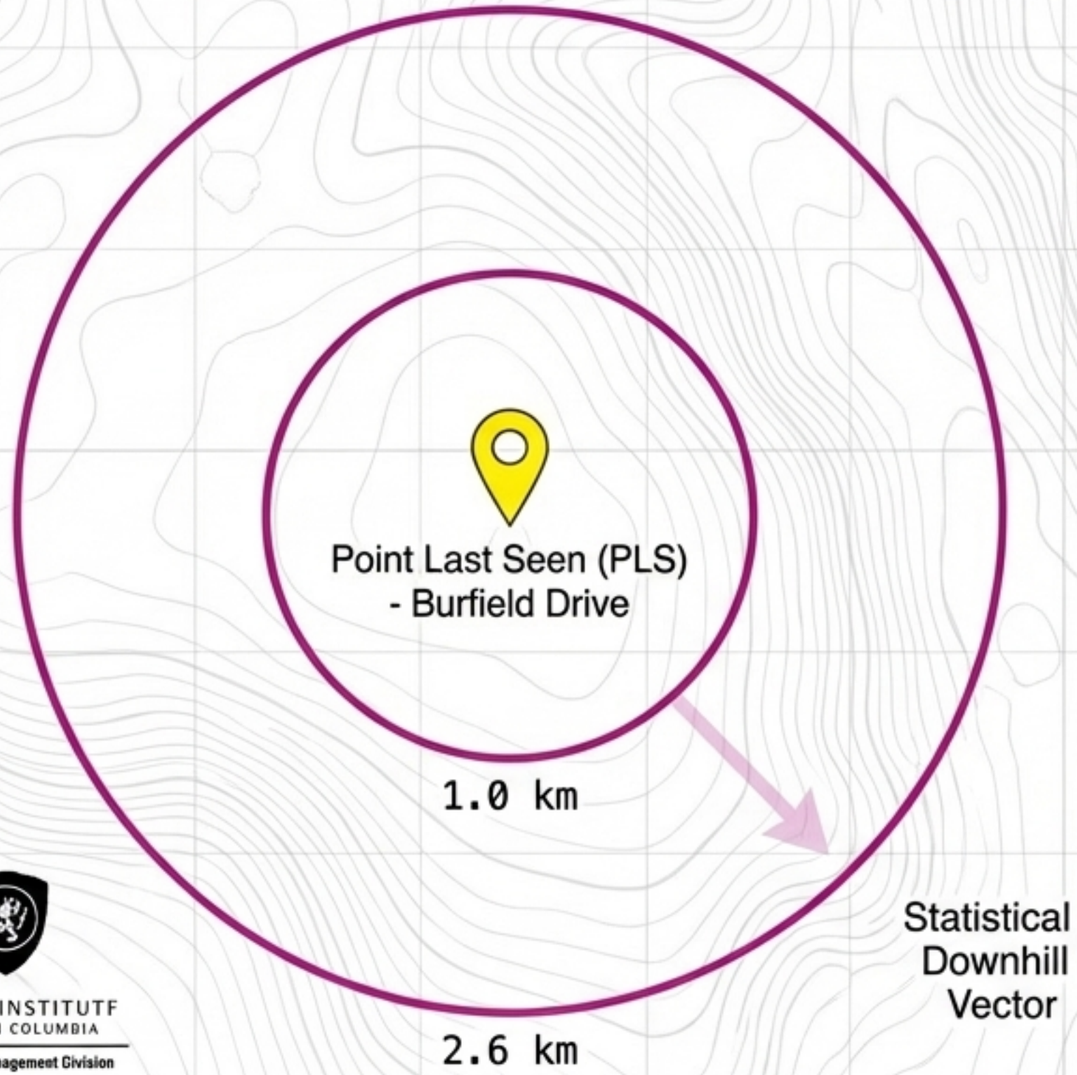
Scans the refined area from the air to penetrate canopy and detect chemical/color signatures or topographical disturbances.

3. Near-Surface Geophysics

Deploys Ground Penetrating Radar (GPR) over high-probability localized anomalies to detect subsurface irregularities without physical disruption.

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Spatial Profiling Parameters



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Point Last Seen (PLS):
House party on Burfield Drive.
Intended destination: 10-15 minute walk.

Trajectory Statistic:
56% to 86% of lost persons head downhill in mountainous terrain.

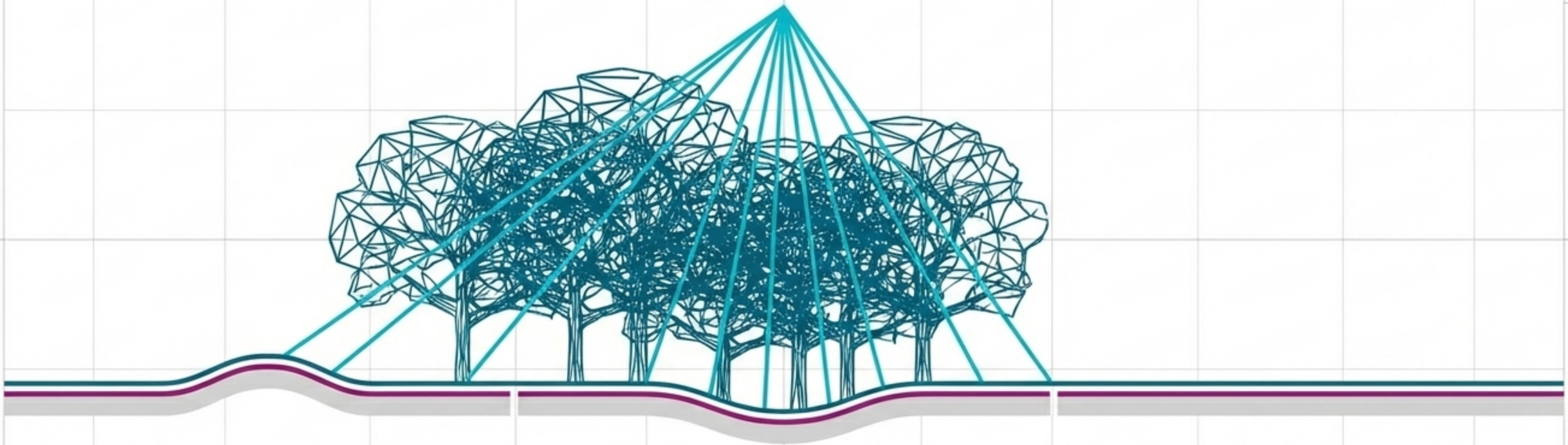
Distance Statistic (Dr. Ken Hill):
The median distance for locating lost persons ranges from 1.0 km to 2.6 km from the PLS.

Forensic Parameter: Primary analytical focus must be placed on downhill trajectories within a 2.6km radius of Burfield Drive.

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Topographical Forensics: LiDAR Elevation Signatures

Technology: Airborne LiDAR uses pulsed, reflected lasers to penetrate dense leaf and forest cover, generating highly precise 3D bare-earth models.



Phase 1: Localized Elevation Gain
Caused by initial disturbance, added mass, or displaced soil/snow.

Phase 2: Localized Elevation Loss
Caused by subsequent soil settling or organic decomposition creating a depression.

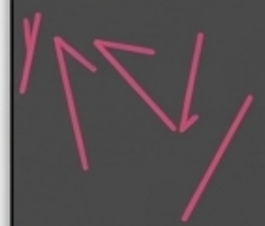
Phase 3: Stasis
Uniform elevation change across all surrounding surfaces.

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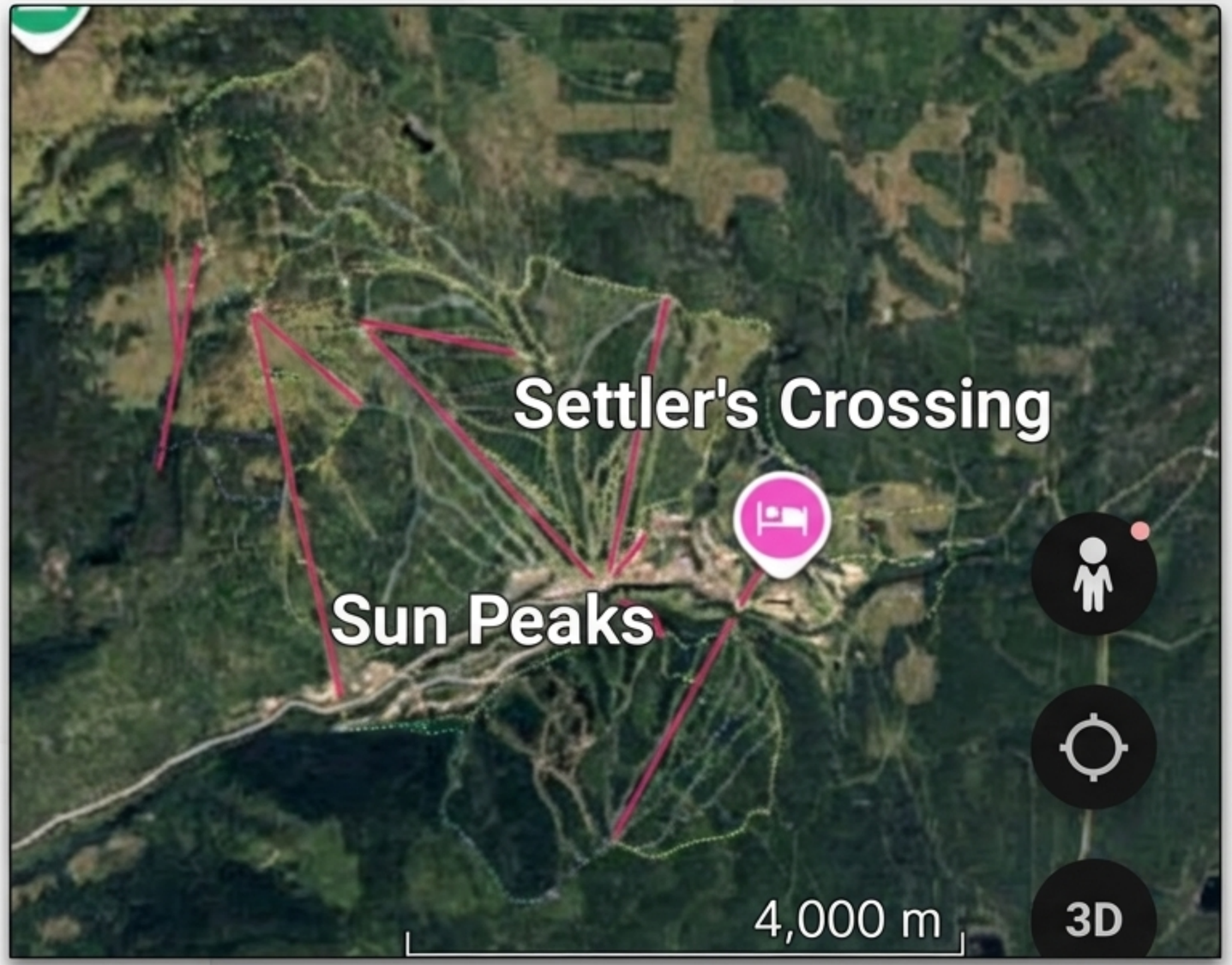
Sector Overview: Sun Peaks Resort Area

Map Coordinates Established:
Overview of the 16 sq km skiable terrain.

Primary Nodes:
Burfield Drive (PLS) and Settler's Crossing (identified by the pink bed icon).

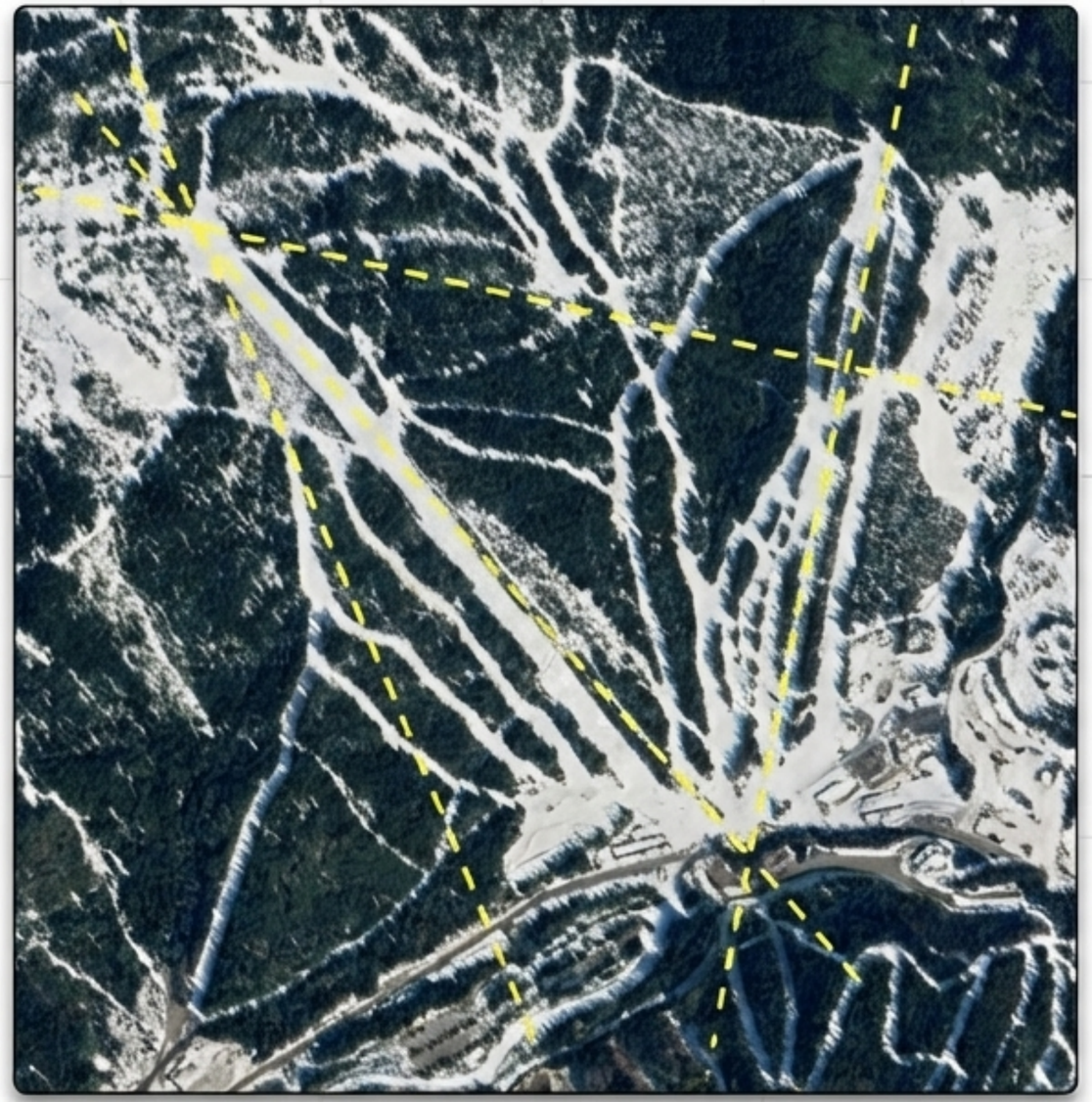


Geometric Zones:
Pink polygonal lines indicate targeted search vectors intersecting both organic wilderness and manmade resort infrastructure.



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Anomaly Classification: Manmade Signatures



- **Visual Profile:** Rigid, uniform, hard-edged geometry.
- **Terrain Features:** Ski runs, clear cuts, logging roads, plowed snowbanks, and structural developments.
- **LiDAR Return:** Shows up as sharp, unnatural breaks in the topographic flow, often with uniform elevation changes due to mechanical grading or backfilling.

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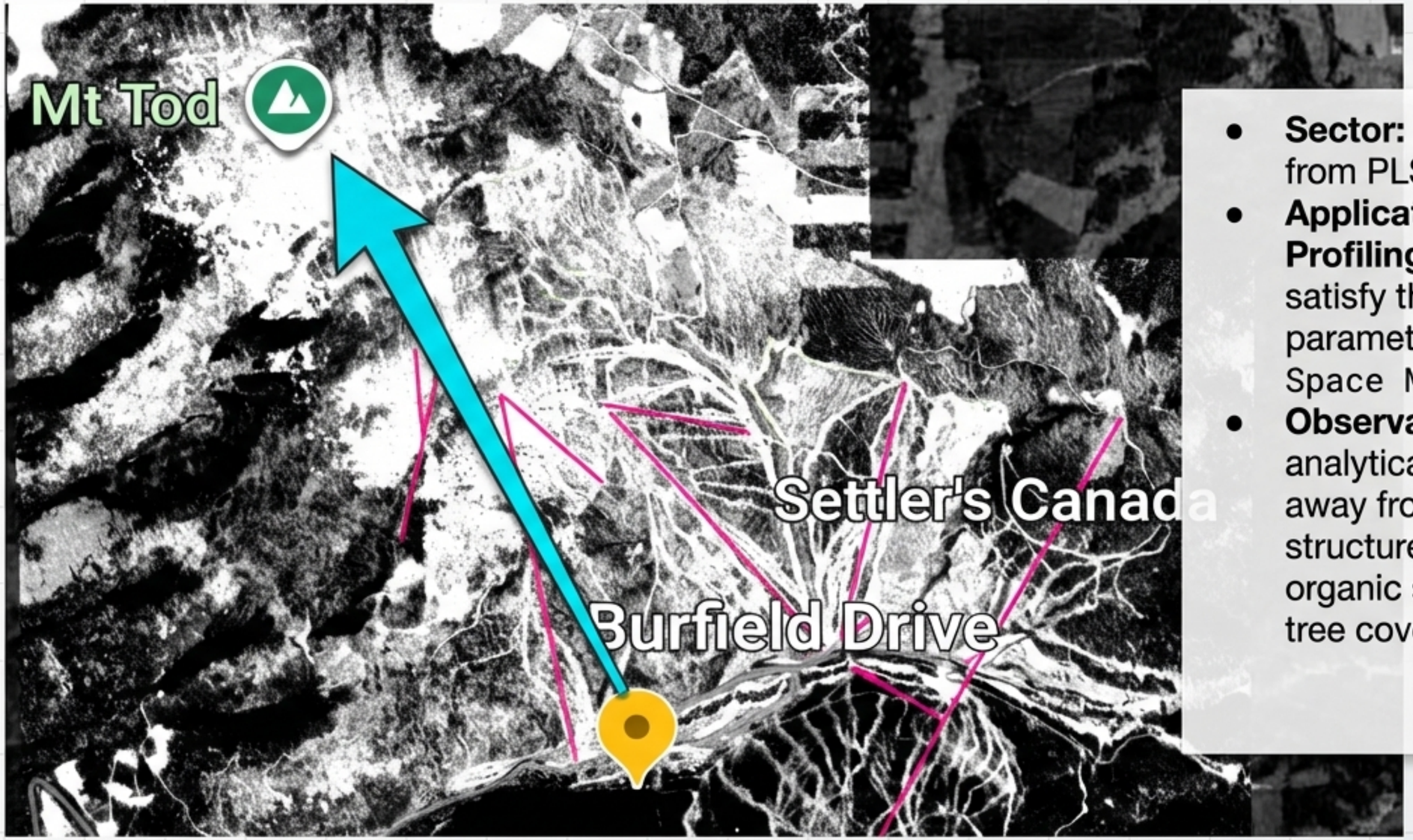
Anomaly Classification: Organic Signatures



- **Visual Profile:** Fractal, irregular, chaotic patterns.
- **Terrain Features:** Steep creek ravines, deep tree wells, dense snow-laden forest canopies.
- **LiDAR Return:** Creates dense ‘noise’ in standard aerial or optical imagery; requires bare-earth LiLiDAR filtering to strip the canopy and reveal the true true ground surface underneath.

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Zone Investigation: Burfield Drive & Mt. Tod



- **Sector:** Northwest trajectory from PLS.
- **Application of Spatial Profiling:** Does this zone satisfy the downhill travel parameter?
  Space Mono
- **Observations:** The pink analytical boundaries track away from the main road structures directly into dense organic signatures (heavy tree cover).

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Zone Investigation: Settler’s Crossing

Sector: East/Southeast of the village core.

Observations: High concentration of manmade edges (plowed roads, property boundaries, ski runs) intersecting with natural forest.

Assessment: Investigating proximity of these vectors to vehicle transit routes.

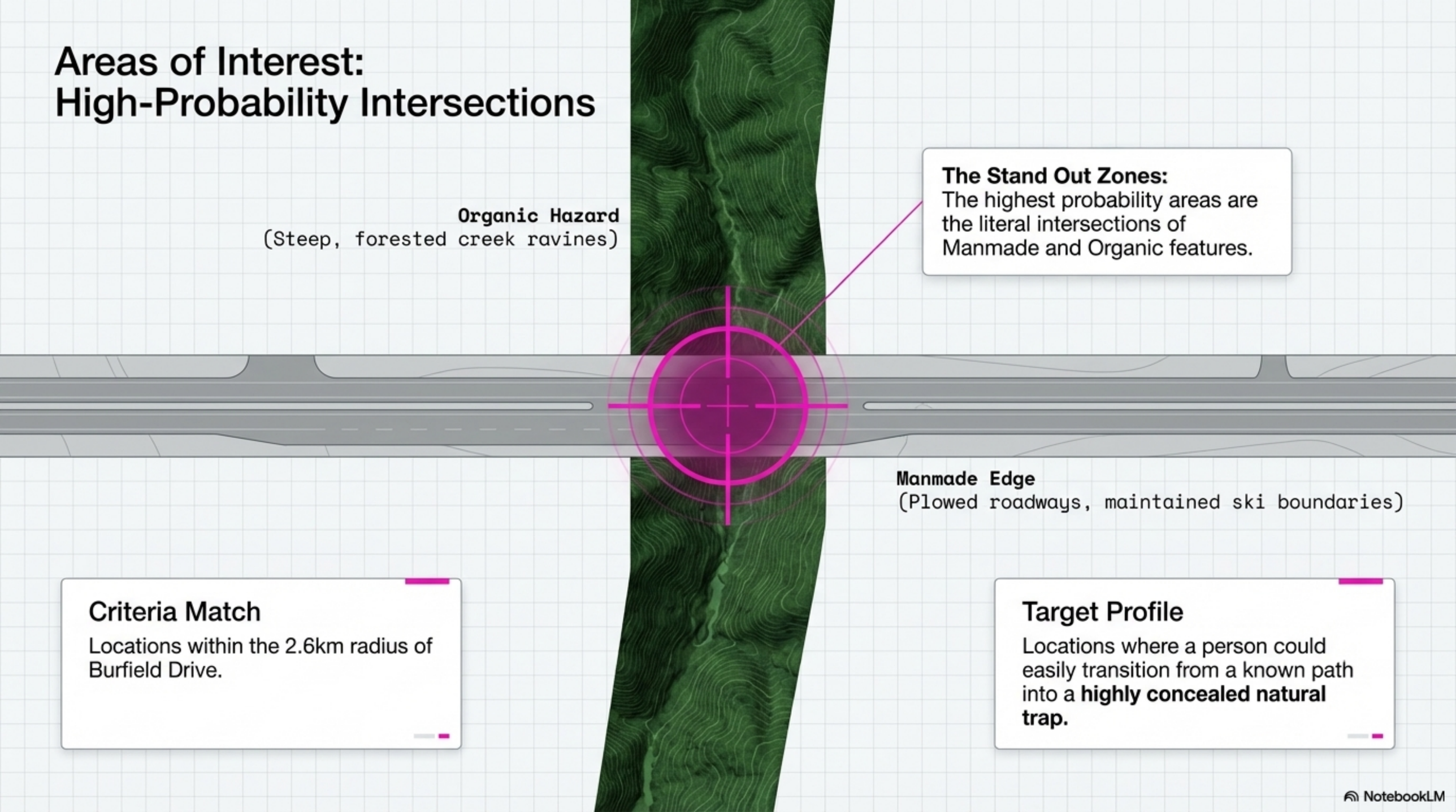


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Areas of Interest: High-Probability Intersections

Organic Hazard
(Steep, forested creek ravines)

The Stand Out Zones:
The highest probability areas are the literal intersections of Manmade and Organic features.



Manmade Edge
(Plowed roadways, maintained ski boundaries)

Criteria Match

Locations within the 2.6km radius of Burfield Drive.

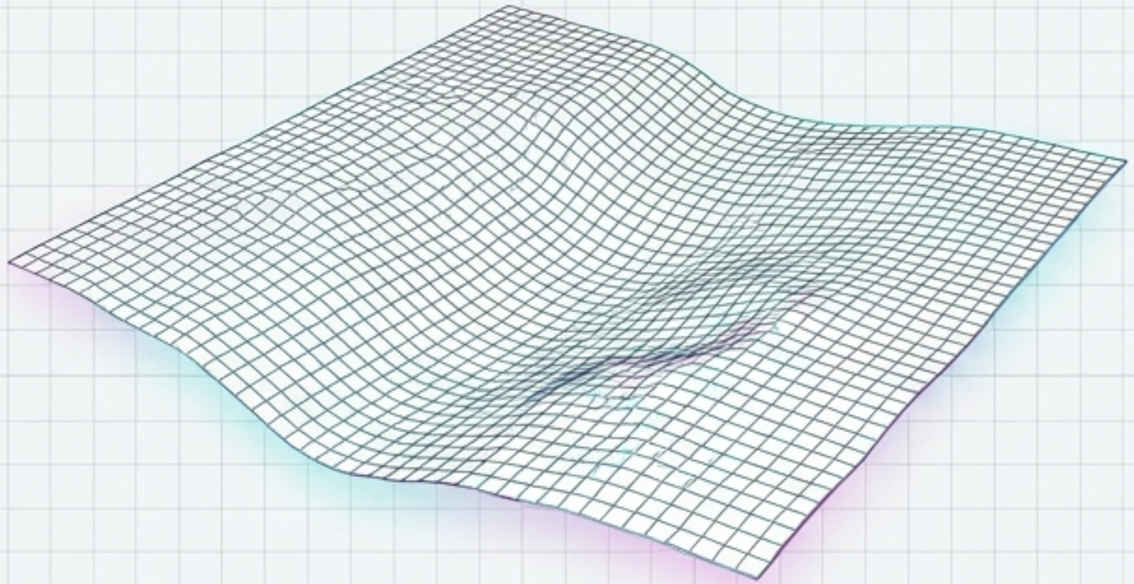
Target Profile

Locations where a person could easily transition from a known path into a **highly concealed natural trap.**

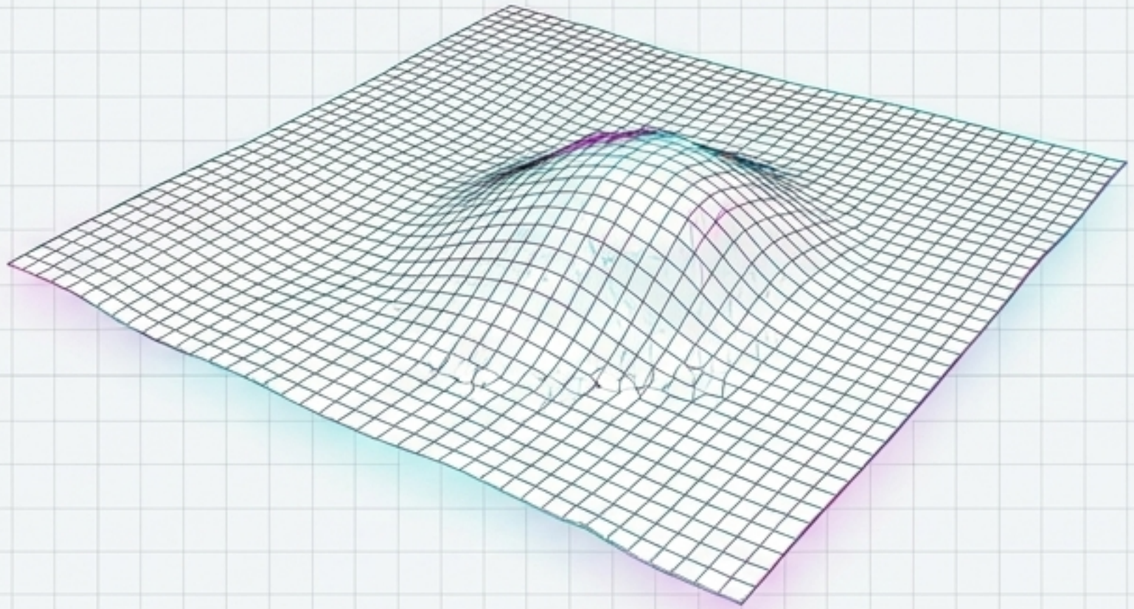
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Analytical Challenge: False Positives

The Resort Complication:
Sun Peaks is an active, heavily maintained 4,400-acre resort.



Natural Terrain Depression



Manmade Snow Pile / Maintenance Debris

- **Mimicry:** Routine trail maintenance, snowplow backfill, and construction debris frequently mimic the localized elevation gains/losses that forensic teams look for.
- **Requirement:** Topographical data must be cross-referenced with resort maintenance logs from **Feb 2018 to eliminate false positives.**

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Technological Deployment: Next Steps



Grid Coverage: 2.6km Radius // Target: Color Signatures

1. Hyperspectral Drone Flights

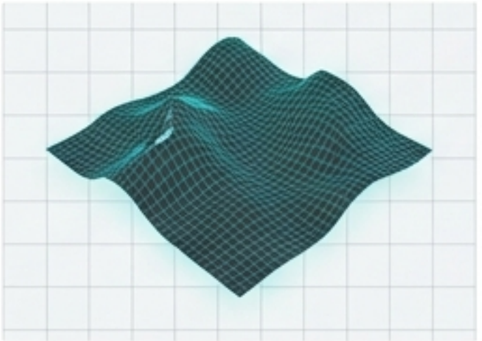
Program autonomous drone grids over the 2.6km radius looking for specific color signatures (e.g., the burgundy ball cap).



Zone Focus: Intersections // Depth Target: Sub-surface Anomalies

2. Localized GPR

Deploy Ground Penetrating Radar over the identified intersection zones (ravines near roads) to map sub-surface anomalies beneath years of accumulated debris.



Data Source: Existing LiDAR // Filter Criteria: Elevation Changes // Reference: Pink Polygon Vectors

3. Refined LiDAR Filtering

Re-process existing LiDAR data specifically isolating elevation changes bordering the pink polygon vectors in Images 1-5.

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Conclusion

Summary: The environment at Sun Peaks is an aggressive adversary, but it is not impenetrable.

Final Thought: By combining the unparalleled perseverance of grassroots ground searches with targeted, military-grade spatial profiling and remote sensing, the rugged terrain can be systematically decoded.

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